

VTU Belagavi

REGULAR EXAMINATION ? 2025

Design and Analysis of Algorithms ? Regular Examination, 2025

Subject Code: CS403 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Asymptotic Analysis)

1. How is Asymptotic Analysis implemented? Give an algorithm / procedure and explain. [12 Marks]
2. Discuss the engineering applications of Asymptotic Analysis. [12 Marks]
3. Derive the expression related to Asymptotic Analysis and explain each term. [12 Marks]

SECTION B (Unit 2: Divide and Conquer)

4. Compare and contrast the key concepts involved in Divide and Conquer. [12 Marks]
5. Write short notes on Divide and Conquer. [12 Marks]
6. Derive the expression related to Divide and Conquer and explain each term. [12 Marks]

SECTION C (Unit 3: Greedy Algorithms)

7. Define Greedy Algorithms. Explain its significance in Design and Analysis of Algorithms with a suitable example. [8 Marks]
8. Define Greedy Algorithms. Explain its significance in Design and Analysis of Algorithms with a suitable example. [12 Marks]
9. Write short notes on Greedy Algorithms. [12 Marks]

SECTION D (Unit 4: Dynamic Programming)

10. How is Dynamic Programming implemented? Give an algorithm / procedure and explain. [12 Marks]
11. Describe the working principle of Dynamic Programming with necessary equations. [12 Marks]
12. Compare and contrast the key concepts involved in Dynamic Programming. [8 Marks]

SECTION E (Unit 5: Graph Algorithms and NP-Completeness)

13. Compare and contrast the key concepts involved in Graph Algorithms and NP-Completeness. [12 Marks]
14. Write short notes on Graph Algorithms and NP-Completeness. [12 Marks]
15. Discuss the engineering applications of Graph Algorithms and NP-Completeness. [12 Marks]

*** End of Question Paper ***